Data Intensive Physics - Project Notes

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Preface

This is a Quarto book.

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1 Data Layout & Specifics

1.1 MER Data Products

```
# Photometric measurements
EUC_MER_FINAL-CAT_TILE102159484-EF6D85_20241026T003047.133877Z_00.00.fits

# Cropped Images of Sources
EUC_MER_FINAL-CUTOUTS-CAT_TILE102159484-73BCF1_20241025T145359.184204Z_00.00.fits

# Galaxy Morphologies
EUC_MER_FINAL-MORPH-CAT_TILE102159484-AE979F_20241026T003045.182961Z_00.00.fits
```

The morphology file contains probabilities for different features, similar to those for the Galaxy zoo project. could be used for a cnn based system??

- The FINAL file contains all photometry information in a large table
- The FINAL-MORPH file contains all morphological measurements
- The FINAL-CUTOUTS file contains the corners for the source cutouts

1.2 SPE Data Products

```
# Spectral line measurements (line name, value, etc for different objects)
EUC_SPE_WIDE-CAT-LIN-102159484_N_20241110T054343.637093Z_00.00.fits

# Parameters required for building spectrum model of each object
EUC_SPE_WIDE-CAT-MOD-102159484_N_20241110T054509.870865Z_00.00.fits

# Photometric redshift estimation
EUC_SPE_WIDE-CAT-Z-102159484_N_20241110T054253.801351Z_00.00.fits
```

- The SPE_WIDE-CAT-MOD file contains four tables:
    1. Catalogue: All spectral lines with: [ID, the central wavelength, the name of the line, the type (absorb or emit)]
    2. Galaxy models: The parameters for a galaxy spectral model for each galaxy candidate
    3. Star models: The parameters for a star spectral model for each star candidate
    4. QSO models: The parameters for a QSO spectral model for each QSO candidate
- The SPE_WIDE-CAT-LIN file contains four tables:
    1. The spectral line measurements
    2. Measurement of absorption atomic indices
    3. Measurement of absorption molecular indices
    4. Continuum features

- The EUC_SPE_WIDE-CAT-Z file contains five tables:
    1. Quality flags
    2. Classification information
    3. Up to 5 redshift candidates for galaxy
    4. 1 relative velocity candidate for star
    5. up to 5 redshift candidates for QSO

1.3 Classifications from OU-PHZ

```
# Spectral Energy Distributions
EUC_PHZ_GALAXYSED__20241120T003959.957136Z_00.00.fits
EUC_PHZ_STARSED__20241120T003936.665995Z_00.00.fits

# Classifications
EUC_PHZ_PHZCAT__20241120T003850.158262Z_00.00.fits
```

- The EUC_PHZ_STARSED file contains the spectral energy distributions for all objects classified as stars
- The EUC_PHZ_GALAXYSED file contains the spectral energy distributions for all objects classified as galaxies
- The EUC_PHZ_PHZCAT file contains the Photometric redshift and probability density function

1.4 Mosaic Images

```
# Mosaic images from different filters
EUC_MER_BGSUB-MOSAIC-NIR-H_TILE102159484-39CF67_20241024T214009.099906Z_00.00.fits
EUC_MER_BGSUB-MOSAIC-NIR-J_TILE102159484-C026DB_20241024T213141.416722Z_00.00.fits
EUC_MER_BGSUB-MOSAIC-NIR-Y_TILE102159484-A9CCA7_20241024T221227.957725Z_00.00.fits
EUC_MER_BGSUB-MOSAIC-VIS_TILE102159484-116033_20241025T015934.885975Z_00.00.fits
```

References